

Processing images in entangled quantum systems

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Abstract We introduce a novel method for storing and retrieving binary geometrical shapes in quantum mechanical systems. In contrast to standard procedures in classical computer science in which image reconstruction requires not only the storage of light parameters (like light frequency) but also the storage and use of additional information like correlation and pixel spatial disposition, we show that the employment of maximally entangled qubits allows to reconstruct images without using any additional information. Moreover, we provide a concrete application of our proposal in the field of image recognition and briefly explore potential experimental realizations. Our proposal could be employed to enable emergent quantum technology to be used in high-impact scientific disciplines in which extensive use of image processing is made.

Keywords Image storage and retrieval · Quantum entanglement · Seevinck–Svetlichny inequalities · Image segmentation

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1 Introduction

The storage, processing and retrieval of digital images are first order tasks for researchers in the discipline of image processing and related areas such as pattern recognition,

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computer vision, and artificial intelligence. However, due to the restricted architecture of classical computers and the often overwhelming computational complexity of state-of-the-art algorithms, it is necessary to find better ways to store, process and retrieve digital images.

In a classical memory device, memory cells are, in terms of hardware, independent of one another, that is, each storage location can be ascribed a reality that is independent of all other locations in the memory. This independence is present in both the formulation of Turing machines as well as in the actual computer manufacturing process. Furthermore, such independence is actually inherited by any kind of information stored in such memory cells. The only way to correlate values stored in such classical computer memories is by means of software, i.e. by either storing (and retrieving) correlation values in memory cells or by executing algorithms on actual numerical values contained in memory cells in order to *calculate* correlation values.

For example, let us follow the process used to store and retrieve the digital photograph shown in Fig. 1. Firstly, each pixel can be seen as a sequence of zeros and ones. Secondly, binary values for each pixel are stored in different and independent memory cells and it is only due to *additional information* regarding the spatial distribution of pixels, also stored in memory cells, that we can reconstruct Lenna's face and the surrounding background in a computer screen. However, if such additional information were lost, reconstructing Lenna's face would be extremely difficult. Moreover, even if information about pixel value and spatial distribution of pixels were available,



Fig. 1 Storing Lenna's face in a digital computer as well as in a hypothetical Turing machine would imply to store data in independent memory cells with no correlation among them a priori. The reconstruction of Lenna's photograph in a computer screen would require additional information with respect to pixel spatial disposition. Lenna's photograph has been widely used by the image processing community for several years

it would not suffice to automatically determine what pixels represent Lenna's eyes, for example. In fact, image segmentation and recognition are open research areas in applied computer science.

Thus, storing an image in a classical (as described above) computer involves a memory which essentially consists of a large set of *independent* bits, each of which represents some property of the associated image (for example and most likely, light frequency or light intensity). Recovery of information concerning the image involves reading binary data stored in the computer memory, that is, an image is recovered by *performing independent measurements of a physical property*, that of electric potential difference, on each cell of the memory device. However, correlations among different points in an image are very important in order to properly understand and describe it. Typically, any one part of an image bears an important relation to other parts. Since the bits that are used to store such images in a classical computer are essentially independent of one another, much relevant information pertaining to the image may be lost upon classical data storage if additional information were not also stored (for example, spatial relation of different constituent elements, like eyes and mouth). Moreover, the reconstruction of images by recovering binary data as described above leads to several subtle but nevertheless important problems in image segmentation and image recognition, e.g. discriminating between different image components with similar colors or gray scales, like dark clothes and walls. Preserving correlation information among pixels could lead to important applications of quantum technology (i.e. quantum algorithms and quantum-mechanical devices) not only in the entertainment industry but also in high-impact scientific fields like the medical sciences. A concrete example could be cancer detection, a scientific discipline in which image analysis is an indispensable tool for modern diagnosis and treatment (see for example [1]).

Let us now consider the storage of a shape through its n vertices in an m -cell memory array. A binary 1 state is stored in a memory cell for each vertex site, while the remaining memory cells are set to zero. If we perform a search of all n vertices with no more information than the 1-0 criterion (i.e. we exhaustively look for n elements contained in a disordered set of cardinality m), it would take $O(m)$ operations to locate such vertices. However, using Grover's algorithm [2] it would only take $O(\sqrt{m})$ operations. Now, in a classical storage array it is not possible to associate vertices with each other in order to identify shapes, without the use of auxiliary information. However, as we shall study in detail in the following section, by exploiting quantum entanglement it is indeed possible to preserve and retrieve information about vertices association without using any auxiliary information. This latter feature is a powerful characteristic as it provides a function that is intensively exploited in many fundamental image processing operations, like edge detection and image segmentation.

In this paper we introduce a novel method for storing and retrieving information about geometrical shapes that does not make any use of auxiliary information and still allows to reconstruct shapes. To do so, our method harnesses a physical property of quantum systems, namely quantum entanglement.

The rest of this paper is divided as follows: in Sect. 2 we present in full detail our novel quantum mechanical method for storing and retrieving geometrical images, Sect. 3 provides a concrete application of our method on image segmentation, Sect. 4 explores some potential realizations of our proposal, and Sect. 5 provides conclusions.

2 New method for storing images

In the discussion that follows, we focus our attention on simple binary images (those images having only two brightness levels, black and white). Such images can be obtained quite simply by thresholding any gray-level image. Whilst restricted in application, such images are of interest because they are relatively straightforward to process and therefore provide a useful starting point for introducing entanglement in the context of image processing.

2.1 Storage of information

We aim to store information concerning the structure and content of a simple image in a quantum system. Consider an array of n qubits which we propose to use as our memory storage. Each qubit in the array may be associated with two parameters, x and y , which together represent grid points of some simple 2D image. Such an array can therefore be used to store visual information. Prior to inputting information into the array, we suppose that each qubit is initialized to state $|0\rangle$. The initial state of the memory is therefore given by the following expression

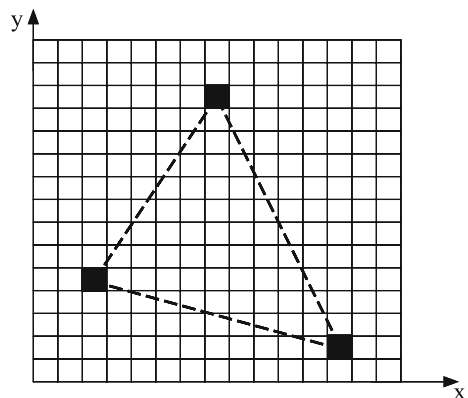
$$|\Psi_{\text{initial}}\rangle = \bigotimes_{i=1}^n |0\rangle_{i(x,y)} \quad (1)$$

We wish to store information about the position and shape of certain simple objects which are represented on our grid as collections of points. Extending the classical binary image formalism to qubits, we associate a white point on the grid with qubit state $|0\rangle$, whilst black corresponds to state $|1\rangle$.

Suppose that we wish to store the shape of a triangle in our qubit array. In this case, we might choose to represent each vertex of the triangle on the grid by setting the corresponding qubit to $|1\rangle$. Such a procedure is depicted in Fig. 2.

The appropriate vertex positions may then be retrieved by applying Grover's quantum search algorithm to the array. We would expect that searching an n -qubit array

Fig. 2 Simple storage procedure for a single triangle in a qubit array. In the classical approach, vertex positions correspond to qubit state $|1\rangle$ and the triangle image can be straightforwardly reconstructed. However, the use of entanglement between vertex locations (*dotted lines*) provides a more fruitful approach



for a $|1\rangle$ using a classical algorithm would take approximately $O(n)$ steps. However, Grover's quantum search algorithm can achieve such a task in approximately $O(\sqrt{n})$ steps. The image of the triangle is then very simply reconstructed from this information.

However, suppose that we instead wish to store *two* triangles in the array. We could proceed as before with an essentially classical approach, preparing the qubits corresponding to triangle vertices in state $|1\rangle$ whilst all others remain in state $|0\rangle$. However, retrieval of information on vertex position by applying Grover's search algorithm will not reveal anything about *which* vertices belong to *which* triangles. We need to store additional information in the array concerning which vertex points belong to which triangle. In this case, entanglement may be employed to establish nonlocal correlations between the qubits storing the vertex locations of the *same* triangle. Consider the maximally-entangled tripartite state [3,4]

$$|\text{GHZ}\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}} \quad (2)$$

Suppose that our qubit array stores a triangle by preparing the associated vertex qubits $\{p, q, r\}$ in a GHZ state. In this case, the memory state of the qubit array is

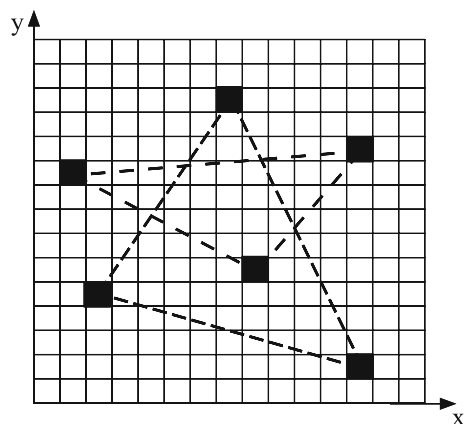
$$|\Psi_1 \text{ triangle}\rangle = \bigotimes_{i=1, i \neq p, q, r}^n |0\rangle_i \otimes \frac{|000\rangle_{pqr} + |111\rangle_{pqr}}{\sqrt{2}} \quad (3)$$

Input of a second triangle with corresponding vertex qubits $\{s, t, u\}$ into the array yields memory state

$$|\Psi_2 \text{ triangles}\rangle = \bigotimes_{i=1, i \neq p, q, r, s, t, u}^n |0\rangle_i \otimes |\text{GHZ}\rangle_{pqr} \otimes |\text{GHZ}\rangle_{stu} \quad (4)$$

Retrieval of the information regarding which particles reside in such maximally entangled states is therefore sufficient to locate the positions of the triangle vertices and also learn to which triangle they belong, as depicted in Fig. 3.

Fig. 3 When two distinct shapes are stored in the array, entanglement (represented by dashed lines) between vertices belonging to the same shape is used to distinguish them from the other



2.2 Retrieval

Once information about an image has been stored in the qubit array it is desirable to retrieve this information in order to reliably reconstruct the image. Information retrieval is achieved by way of performing measurements on the array.

Suppose that we store a triangle in the array in the form of the state $|\text{GHZ}\rangle_{pqr}$. Information pertaining to relations between one memory location for the image and another could be retrieved from the array by implementing the measurement projection operator

$$\hat{M}_{pqr} = |00\dots 0\rangle |\text{GHZ}\rangle_{pqr} \langle \text{GHZ}| \langle 00\dots 0| \quad (5)$$

where $|00\dots 0\rangle$ acts on all qubits not belonging to the GHZ state. However, since any GHZ state consists of a coherent superposition of $|000\rangle$ and $|111\rangle$, the qubit array has the form of a coherent superposition

$$\begin{aligned} |\Psi_{\text{triangle}}\rangle &= \frac{1}{\sqrt{2}} \left[\bigotimes_{i=1}^n |0\rangle + \bigotimes_{i \neq p,q,r}^n |0\rangle \otimes |111\rangle_{pqr} \right] \\ &= \frac{1}{\sqrt{2}} \left[|\Psi_{\text{initial}}\rangle + \bigotimes_{i \neq p,q,r}^n |0\rangle_i \otimes |111\rangle_{pqr} \right] \end{aligned}$$

In fact, any memory state of the qubit array will consist of a coherent superposition of $|\Psi_{\text{initial}}\rangle$ and other memory states. Therefore memory states associated with different images are nonorthogonal and cannot be distinguished unambiguously. This means that using projection operators will only give probabilistic results for vertex location and the image cannot be reliably reconstructed. Instead, a measurement probing the *entanglement* shared between the vertex qubits is employed in order to determine their location.

Bell-type inequalities [5] for N -particle systems have been derived under the assumption of total and partial separability and provide us with a way of deciding whether a set of N particles resides in a *maximally entangled state*. We have chosen the Seevinck–Svetlichny inequalities [6, 7] for building our image retrieval protocol, as such inequalities have been shown to be maximally violated by N -particle GHZ states.

In particular, a partially-entangled tripartite state may be written in the form

$$\hat{\varrho}^{(123)} = p_r \hat{\varrho}^{(12)} \otimes \hat{\varrho}^{(3)} + p_s \hat{\varrho}^{(13)} \otimes \hat{\varrho}^{(2)} + p_t \hat{\varrho}^{(23)} \otimes \hat{\varrho}^{(1)} \quad (6)$$

where the $\hat{\varrho}^{(ij)}$ represent entangled states of two of the subsystems involved. Any state $\hat{\varrho}^{(123)}$ that obeys $\hat{\varrho}^{(123)} \neq \sum_i p_i \hat{\varrho}_i$, where all the $\hat{\varrho}_i$ are separable into products of states of less than three parties, is fully entangled.

To search for triangles, a set of three qubits in the array is chosen for measurement. This tripartite state is then tested for violation of the Seevinck–Svetlichny inequalities for tripartite states. Violation implies the presence of full three-particle entanglement. Non-violation therefore implies that the three qubits selected *do not* form vertices of

the same triangle. From this information it is straightforward to deduce the location of the triangle vertices. Now suppose that the three qubits selected consist of two qubits residing in the *same* GHZ state and a third that does not. Then the state to be tested is of the form presented in Eq. (6) and will not violate the appropriate inequalities for *full* tripartite entanglement. For our simple example of two triangles, determinations of the locations of all six vertices requires at worst ${}^nC_3 \times {}^{n-3}C_3$ different identically-prepared arrays to be tested for two instances of tripartite entanglement amongst different qubits. Indeed, in exchange for not using software and due to the statistical nature of quantum mechanics, several quantum hardware grids must be identically prepared. However, quantum technology realizations may eventually become cheap enough to allow for massive use in high-impact technology.

Finally, the rationale followed above may be straightforwardly used for storing and retrieving more complex polygons of N sides, as it would suffice to use N -particle GHZ states for storing each polygon, and to apply the Seevinck–Svetlichny inequalities [6, 7] for polygon retrieval.

3 Application: image segmentation using maximally entangled states

Image segmentation, an active research field in computer vision and image processing, is defined as the process of dividing an image into separate regions [8]. For example, we may be interested in segmenting out all faces from a birthday party picture, or all compact vehicles from a picture of a traffic jam.

Clearly, the process of segmenting out a picture is context-dependent, i.e. our choice of pixels from an image or a set of images primarily depends on the purpose that lies behind our selection criterion. For example, choosing faces from a birthday party picture could be driven by our wish to see again our beloved ones, while selecting all compact vehicles may be performed by a police looking for a felon.

Thus, performing image segmentation by automatic means, i.e. by means of computers and algorithms, is a challenging task not only due to the technical aspects of determining whether a pixel belongs to a certain set or not, but also because we must find ways to program a computer, i.e. to instruct it, in order to employ the selection criteria we have designed.

There are several methods for performing image segmentation [8, 9]: we may use a technique known as feature detection which consists of detecting abrupt changes in pixel intensity while moving a small window around an image; another method consists of employing the eigenvalues of a matrix whose entries are functions of pixel intensities; also, if a sequence of images of the same moving target is available, we may be able to automatically segment out such a target by computing (numerical) partial derivatives on pixel intensities. Finally, we may look for the edges of an object in an image by taking a few interior points of such image and iteratively growing the covered area, the stop criterion being an abrupt change in pixel intensity in a small neighborhood. This last method may use, for iteratively increasing the covered area, operators from mathematical morphology [10].

Techniques for image segmentation may work for some images and may not work for others, depending on the properties of the particular image under study. A concrete



Fig. 4 Traditional image segmentation techniques fail mainly due to the complexities associated to pixel intensity and spatial location of objects in an image, as well as to the fact that information about objects and structure in an image is seldom provided before executing image segmentation algorithms. In contrast, humans are tremendously good at recognizing objects because we have information stored in our brains and we can use that information to determine what objects an image is composed of. Concretely, we use our knowledge and intuition to rapidly infer which pixels belong to which leaves in the picture shown in this figure

example can be seen in Fig. 4 in which we show a picture of a bush. All segmentation techniques introduced in the previous paragraph would fail if employed to detect each and every single leaf in this picture. For example, abrupt changes in pixel intensity may not be detected in contiguous leaves with similar grayscale values and somewhat blurred edges; moreover, edges could be missed in overlapping leaves. In many cases, image segmentation techniques fail due to the fact that segmentation is performed without any previous information about what to expect in an image. In contrast, human beings are extremely good at image segmentation partly due to the information we acquire as we grow up as well as to the learning and inference capabilities developed by human brains.

In this section we shall explore how our novel method for storing, processing and retrieving an image using maximally entangled quantum systems may be used in image segmentation algorithms.

3.1 Segmenting non-overlapping an overlapping elements

Let us suppose we use a multipartite quantum memory as the one described in Sect. 2 to store a black and white representation of images shown in Fig. 5. We can see that pictures (a) and (b) from Fig. 5 are the same except for the objects we have segmented out. In Fig. 5a we have selected all chairs found in the living room (borders have been stressed using white thin lines) while in Fig. 5b we have chosen all framed pictures found in the living room (borders have been stressed using dark gray thick lines).

Suppose we use a qubit grid as the one shown in Fig. 2 to store images (a) and (b) from Fig. 5. We could then employ maximally entangled quantum states (e.g. states described by Eq. 2) to store the vertices of geometrical objects (i.e. polygons) that correspond to the corners of each chair and each frame. We would be in a case similar to that described in Fig. 2, i.e. we would use quantum entanglement to store information

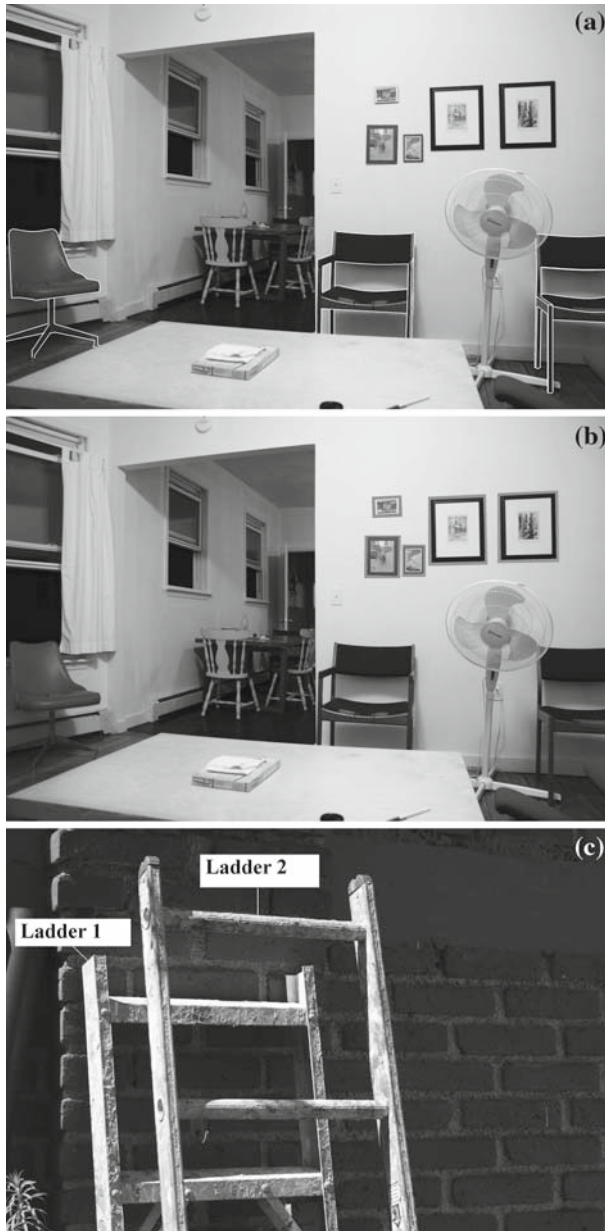


Fig. 5 Using maximally entangled states for image segmentation. As for non-overlapping objects, we use maximally entangled states to store points that correspond to corners of each chair shape in **a** and each frame in **b**. At a later stage, we could then detect quantum entanglement using the technique developed in Sect. 2.2 to determine which pixels belong to which objects. Overlapping objects could be segmented out using a similar rationale. For example, the corners of both ladders from **c** could be mapped into maximally entangled states. At a later stage, qubits corresponding to those maximally entangled states would be retrieved and this information could be straightforwardly used to determine which pixels belong to which ladder

about points that belong to (non-overlapping) objects we want to segment out, that is, points whose locations are correlated due to the segmentation criterion we have chosen.

A concrete application of the set-up described above follows: assume we have stored, in maximally entangled states, the corners of each chair shape from Fig. 5a. After detection of maximally entangled states using the technique provided in Sect. 2.2, we would know *which vertices correspond to which chairs*. Then, we could use those detected corners to build polygons that correspond to chair shapes and, finally, we could use mathematical morphology operators to fill out the space bordered by those polygons. As a result, we would know which pixels belong to which chairs, making the segmentation of chairs from the image under study a straightforward procedure.

A similar procedure could be employed to segment out overlapping images from a picture, as corners from each overlapping image could be mapped into a maximally entangled state, as in Fig. 3. After detection of those maximally entangled states we may use them to reconstruct the shapes that correspond to polygons drawn by linking corresponding corners. A concrete example is provided in Fig. 5c.

4 Exploring potential implementations

Building experimental multipartite quantum-entangled systems is a challenging task, particularly due to the difficulty of scaling realizations on the number of qubits [11]. Thus, in this section we briefly describe five experimental realizations of quantum-mechanical systems that, in principle, could be employed to implement our image processing method based on quantum entanglement, bearing in mind that a definite platform for such implementation will largely depend on future experimental results on N-particle GHZ states generation.

Zou and Mathis [12] have proposed using three electronic levels of several Rydberg atoms in microwave cavity QED to implement, in one step, GHZ states. Moreover, Leibfried et al. [13] have shown that it is possible to create 6-qubit 'Schrödinger cat' states using the same number of beryllium ions, each qubit being defined by two hyperfine ground states of each beryllium atom.

Another plausible implementation of our proposal has been developed by Wei et al. [14], where superconducting qubits have been used to build GHZ states (in this case, qubits are defined by charge Cooper-pair boxes). Also, Walther et al. [15] have presented a scheme, based on photon polarization, for generating N-particle GHZ states. Finally, Fiorentino and Wong [16] have proposed a scheme, based on hyper-entangled states [17] and linear optics, in which hyperentangled photons (entangled degrees of freedom are photon polarization and momentum) can be used to build GHZ states.

5 Conclusions

We have developed a novel method for storage and retrieval of simple binary images in quantum systems. Our method does not make any use of auxiliary information and still allows to reconstruct shapes due to the employment of quantum entanglement. Moreover, we have presented a proof of concept about the plausibility of using

our proposal in image segmentation, and have briefly elaborated on several potential experimental implementations. Our proposal could be employed in applications of quantum technology in which preserving correlations found in observed systems in actual quantum hardware (instead of hardware and software) would be advantageous.

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